Azure NetApp Files documentation

Azure NetApp Files

NetApp
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Azure NetApp Files documentation

Release notes

What's new

Learn what's new with Azure NetApp Files in BlueXP.

11 April 2021

Support for volume templates

A new Application Templates service enables you to set up a volume template for Azure NetApp Files. The template should make your job easier because certain volume parameters will already be defined in the template, such as capacity pool, size, protocol, VNet and subnet where the volume should reside, and more. When a parameter is already predefined, you can just skip to the next volume parameter.

- Learn about Application Templates and how you can use them in your environment
- Learn how to create an Azure NetApp Files volume from a template

8 March 2021

Dynamically change service levels

You can now dynamically change the service level for a volume to meet workload needs and optimize your costs. The volume is moved to the other capacity pool with no impact to the volume.

Learn how to change a volume's service level.

3 August 2020

Azure NetApp Files set up and management

Set up and manage Azure NetApp Files directly from Cloud Manager. After you create an Azure NetApp Files working environment, you can complete the following tasks:

- Create NFS and SMB volumes.
- Manage capacity pools and volume snapshots

  Cloud Manager enables you to create, delete, and restore volume snapshots. You can also create new capacity pools and specify their service levels.

- Edit a volume by changing its size and managing tags.

The ability to create and manage Azure NetApp Files directly from Cloud Manager replaces the previous data migration functionality.

5 April 2020

Data migration to Azure NetApp Files

You can now migrate NFS or SMB data to Azure NetApp Files directly from Cloud Manager. Data syncs are powered by BlueXP copy and sync.

Get started

Learn about Azure NetApp Files

Azure NetApp Files enables enterprises to migrate and run their performance-intensive and latency-sensitive core, business-critical applications in Azure with no need to refactor for the cloud.

Features

- Support for multiple protocols enables "lift & shift" of both Linux & Windows applications to run seamlessly in Azure.
- Multiple performance tiers allow for close alignment with workload performance requirements.
- Leading certifications including SAP HANA, GDPR, and HIPAA enables migration of the most demanding workloads to Azure.

Additional features in BlueXP

- Migrate NFS or SMB data to Azure NetApp Files directly from BlueXP. Data migrations are powered by BlueXP copy and sync.

  Learn more about BlueXP copy and sync

- Using Artificial Intelligence (AI) driven technology, BlueXP classification can help you understand data context and identify sensitive data that resides in your Azure NetApp Files accounts.

  Learn more about BlueXP classification

Cost

View Azure NetApp Files pricing

Note that your subscription and charging are maintained by the Azure NetApp Files service and not by BlueXP.

Supported regions

View supported Azure regions

Getting help

For technical support issues associated with Azure NetApp Files, use the Azure portal to log a support request to Microsoft. Select your associated Microsoft subscription and select the **Azure NetApp Files** service name under **Storage**. Provide the remaining information required to create your Microsoft support request.

Related links

- NetApp BlueXP website: Azure NetApp Files
- Azure NetApp Files documentation
- BlueXP copy and sync documentation

Getting started workflow

Get started with Azure NetApp Files by setting up a Microsoft Entra application and by creating a working environment.

1 **Set up a Microsoft Entra application**

From Azure, grant permissions to a Microsoft Entra application and copy the application (client) ID, the directory (tenant) ID, and the value of a client secret.

2 **Create an Azure NetApp Files working environment**

From BlueXP, click **Add Working Environment** > **Microsoft Azure** > **Azure NetApp Files** and then provide details about the AD application.

Set up a Microsoft Entra application

BlueXP needs permissions to set up and manage Azure NetApp Files. You can grant the required permissions to an Azure account by creating and setting up a Microsoft Entra application and by obtaining the Azure credentials that BlueXP needs.

Step 1: Create the application

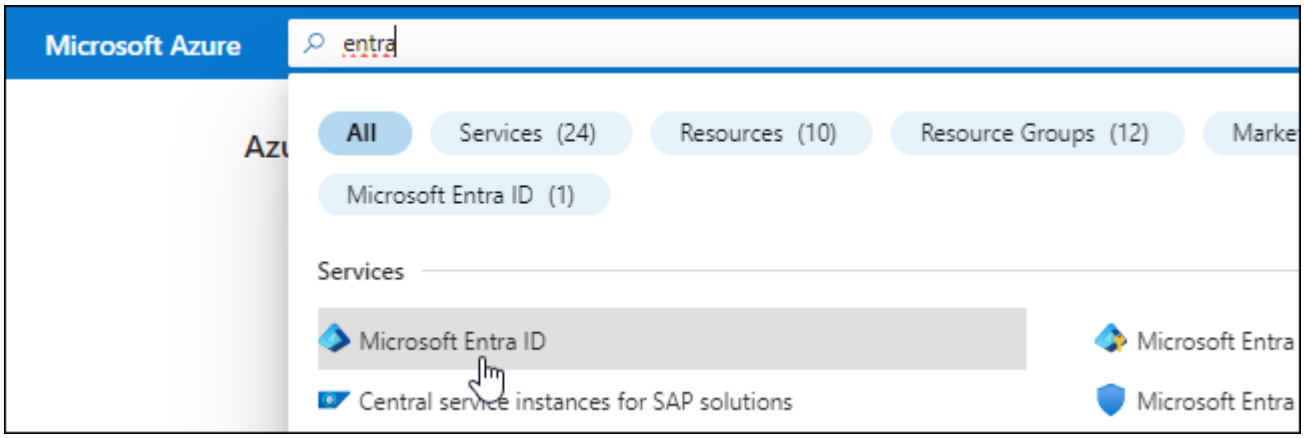
Create a Microsoft Entra application and service principal that BlueXP can use for role-based access control.

Before you begin

You must have the right permissions in Azure to create an Active Directory application and to assign the application to a role. For details, refer to Microsoft Azure Documentation: Required permissions.

Steps

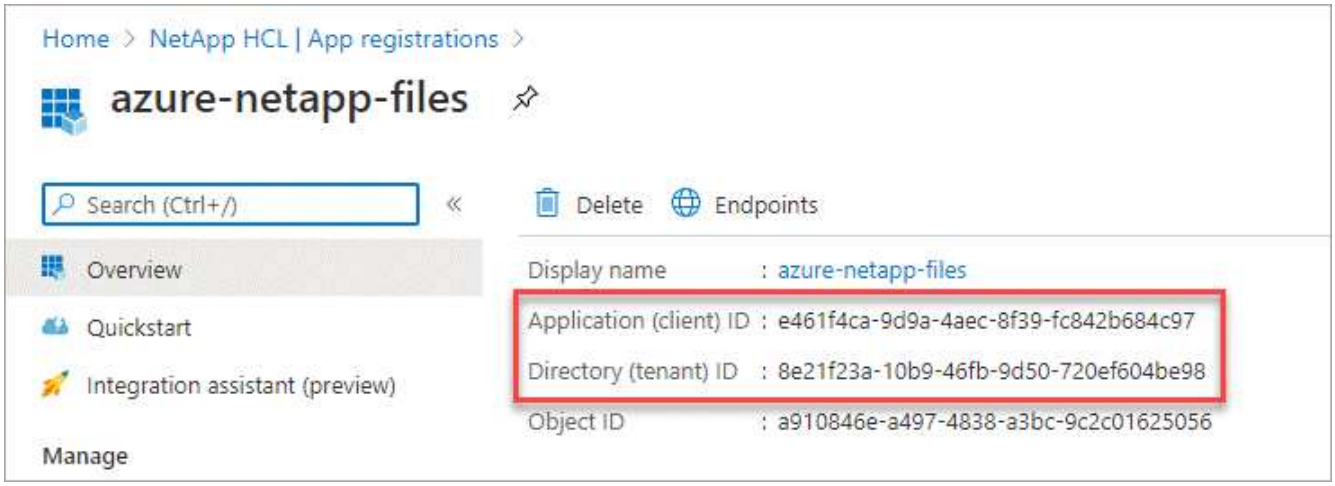
1. From the Azure portal, open the **Microsoft Entra ID** service.



2. In the menu, click **App registrations**.
3. Create the application:
   a. Click **New registration**.

b. Specify details about the application:
   - **Name**: Enter a name for the application.
   - **Account type**: Select an account type (any will work with BlueXP).
   - **Redirect URI**: You can leave this blank.
c. Click **Register**.

4. Copy the **Application (client) ID** and the **Directory (tenant) ID**.



When you create the Azure NetApp Files working environment in BlueXP, you need to provide the application (client) ID and the directory (tenant) ID for the application. BlueXP uses the IDs to programmatically sign in.

5. Create a client secret for the application so BlueXP can use it to authenticate with Microsoft Entra ID:
   a. Click **Certificates & secrets > New client secret**.
   b. Provide a description of the secret and a duration.
   c. Click **Add**.
   d. Copy the value of the client secret.

Result

Your AD application is now setup and you should have copied the application (client) ID, the directory (tenant) ID, and the value of the client secret. You need to enter this information in BlueXP when you add an Azure NetApp Files working environment.

Step 2: Assign the app to a role

You must bind the service principal to your Azure subscription and assign it a custom role that has the required permissions.

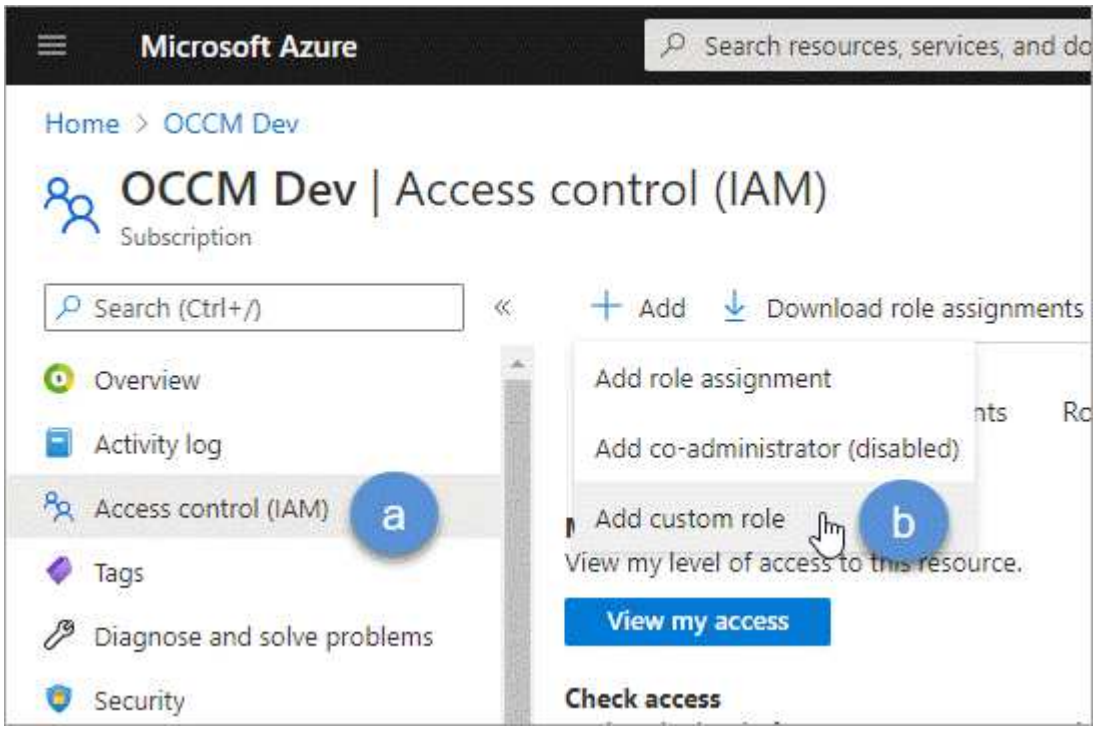
Steps

1. Create a custom role in Azure.

   The following steps describe how to create the role from the Azure portal.

   a. Open the subscription and click **Access control (IAM)**.

   b. Click **Add > Add custom role**.



   c. In the **Basics** tab, enter a name and description for the role.

   d. Click **JSON** and click **Edit** which appears at the top right of the JSON format.

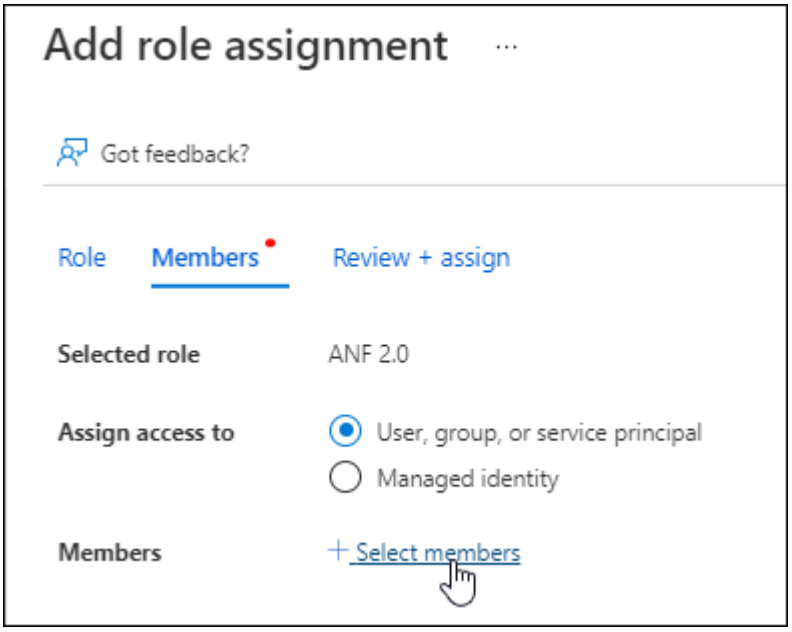
   e. Add the following permissions under *actions*:

```
"actions": [
    "Microsoft.NetApp/*",
    "Microsoft.Resources/resources/read",
    "Microsoft.Resources/subscriptions/resourceGroups/read",

"Microsoft.Resources/subscriptions/resourcegroups/resources/read",
    "Microsoft.Resources/subscriptions/resourceGroups/write",
    "Microsoft.Network/virtualNetworks/read",
    "Microsoft.Insights/Metrics/Read"
    ],
```

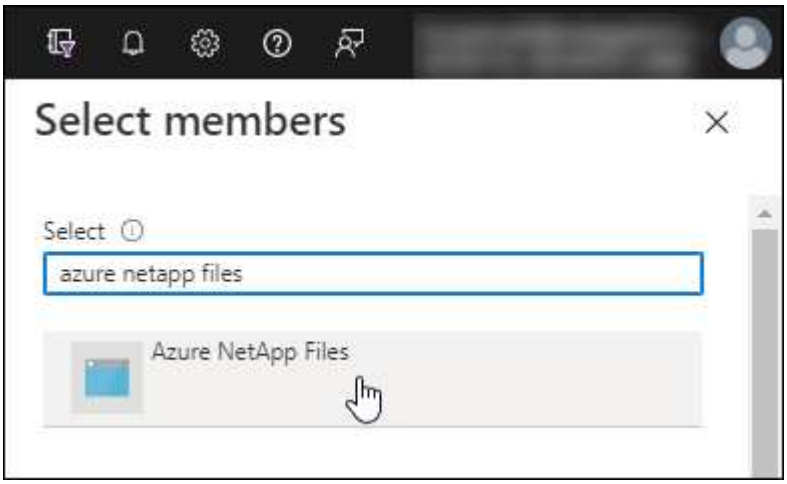
f. Click **Save**, click **Next**, and then click **Create**.

2. Now assign the application to the role that you just created:
   a. From the Azure portal, open the **Subscriptions** service.
   b. Select the subscription.
   c. Click **Access control (IAM) > Add > Add role assignment**.
   d. In the **Role** tab, select the custom role that you created and click **Next**.
   e. In the **Members** tab, complete the following steps:
      - Keep **User, group, or service principal** selected.
      - Click **Select members**.



      - Search for the name of the application.

        Here's an example:



      - Select the application and click **Select**.
      - Click **Next**.

f. Click **Review + assign**.

The service principal for BlueXP now has the required Azure permissions for that subscription.

Step 3: Add the credentials to BlueXP

When you create the Azure NetApp Files working environment, you're prompted to select the credentials associated with the service principal. You need to add these credentials to BlueXP before you create the working environment.

Steps

1. In the upper right of the BlueXP console, click the Settings icon, and select **Credentials**.

2. Click **Add Credentials** and follow the steps in the wizard.
   a. **Credentials Location**: Select **Microsoft Azure > BlueXP**.
   b. **Define Credentials**: Enter information about the Microsoft Entra service principal that grants the required permissions:
      - Client Secret
      - Application (client) ID
      - Directory (tenant) ID

      You should have captured this information when you created the AD application.

   c. **Review**: Confirm the details about the new credentials and click **Add**.

Create an Azure NetApp Files working environment

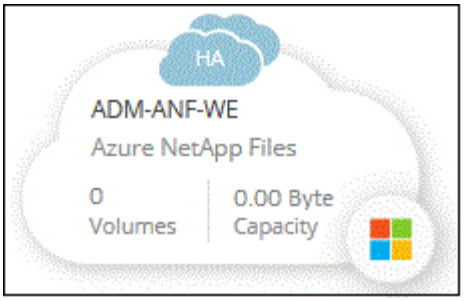
After you set up a Microsoft Entra application and add the credentials to BlueXP, create an Azure NetApp Files working environment so that you can start creating the volumes that you need.

Steps

1. From the Canvas page, click **Add Working Environment**.
2. Select **Microsoft Azure**.
3. Next to Azure NetApp Files, select **Discover**.
4. On the Details page, enter a working environment name and select the credentials that you previously set up.
5. Click **Continue**.

Result

You should now have an Azure NetApp Files working environment.



What's next?

Start creating and managing volumes.

Use Azure NetApp Files

Create and mount volumes

After you set up your working environment, you can create Azure NetApp Files accounts, capacity pools, and volumes.

Create volumes

You can create NFS or SMB volumes in a new or existing Azure NetApp Files account.

A BlueXP feature called "templates" enables you to create volumes that are optimized for the workload requirements for certain applications; such as databases or streaming services. If your organization has created volume templates that you should use, follow these steps.

Before you begin

- If you want to use SMB, you must have set up DNS and Active Directory.
- When planning to create an SMB volume, you must have a Windows Active Directory server available to which you can connect. You will enter this information when creating the volume.

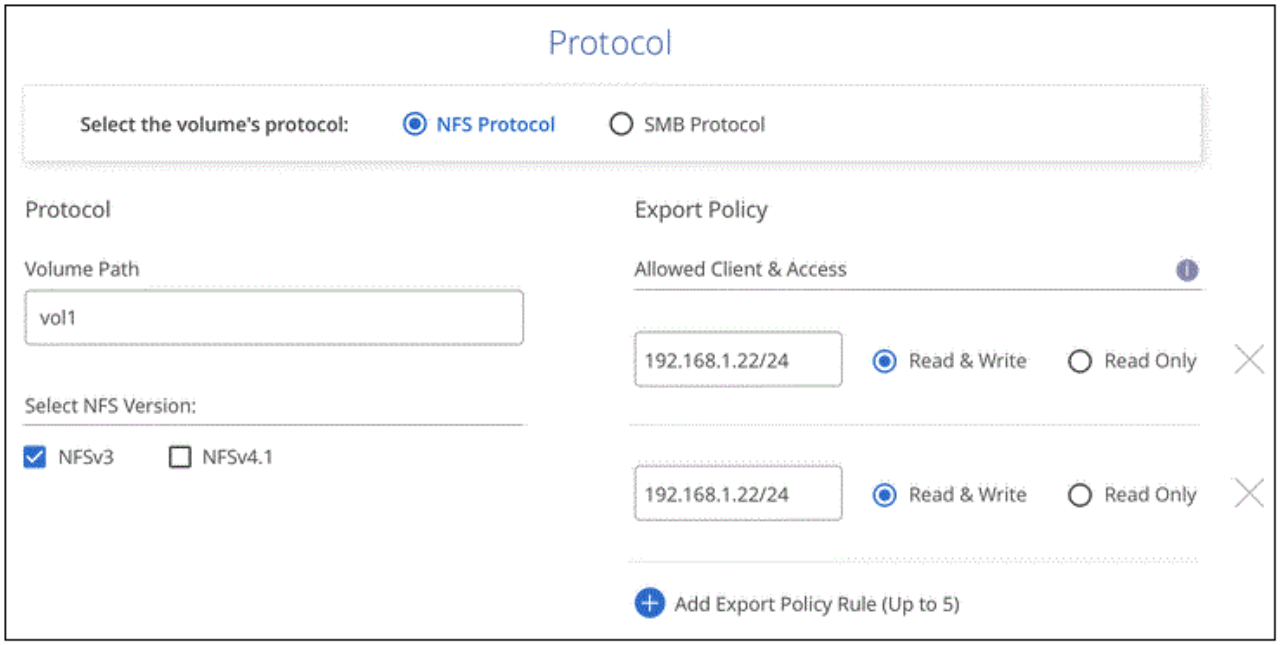
Steps

1. Open the Azure NetApp Files working environment.
2. Click **Add New Volume**.
3. Provide the required information on each page:
    - **Azure NetApp Files Account**: Choose an existing Azure NetApp Files account or create a new account. When creating a new account you can also choose the Resource Group that you want to use.
    - **Capacity Pool**: Select an existing capacity pool or create a new capacity pool.

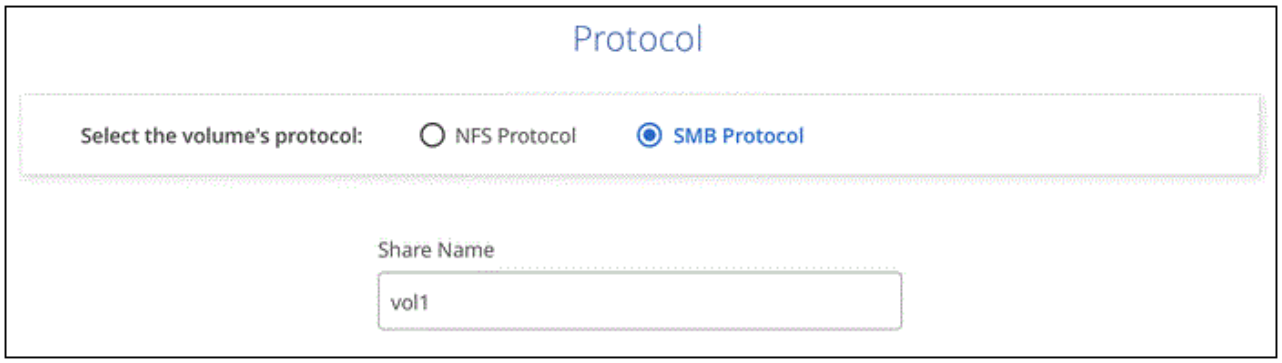
      If you create a new capacity pool, you need to specify a size and select a service level.

      The minimum size for the capacity pool is 4 TB. You can specify a size in multiples of 4 TB.
    - **Details & Tags**: Enter a volume name and size, the VNet and subnet where the volume should reside, and optionally specify tags for the volume.
    - **Protocol**: Choose the NFS or SMB protocol and enter the required information.

      Here’s an example of details for NFS.



Here's an example of details for SMB. You'll need to provide Active Directory information on the next page when you set up your first SMB volume.



4. If you want this volume to be created based on a snapshot of an existing volume, select the snapshot from the Snapshot Name drop-down list.
5. Click **Add Volume**.

Result

The new volume is added to the working environment.

Continue with mounting the cloud volume.

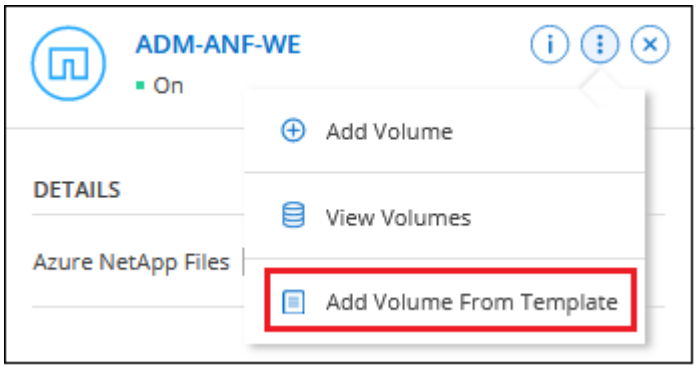
Create volumes from templates

If your organization has created Azure NetApp Files volume templates so that you can deploy volumes that are optimized for the workload requirements for certain applications, follow the steps in this section.

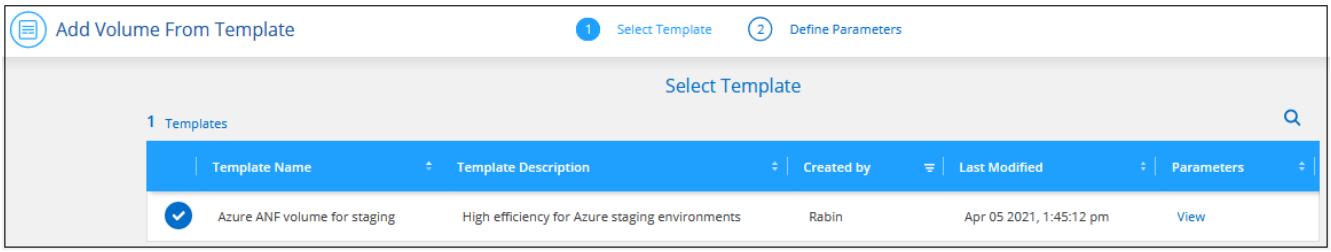
The template should make your job easier because certain volume parameters will already be defined in the template, such as capacity pool, size, protocol, VNet and subnet where the volume should reside, and more. When a parameter is already predefined, you can just skip to the next volume parameter.

Steps

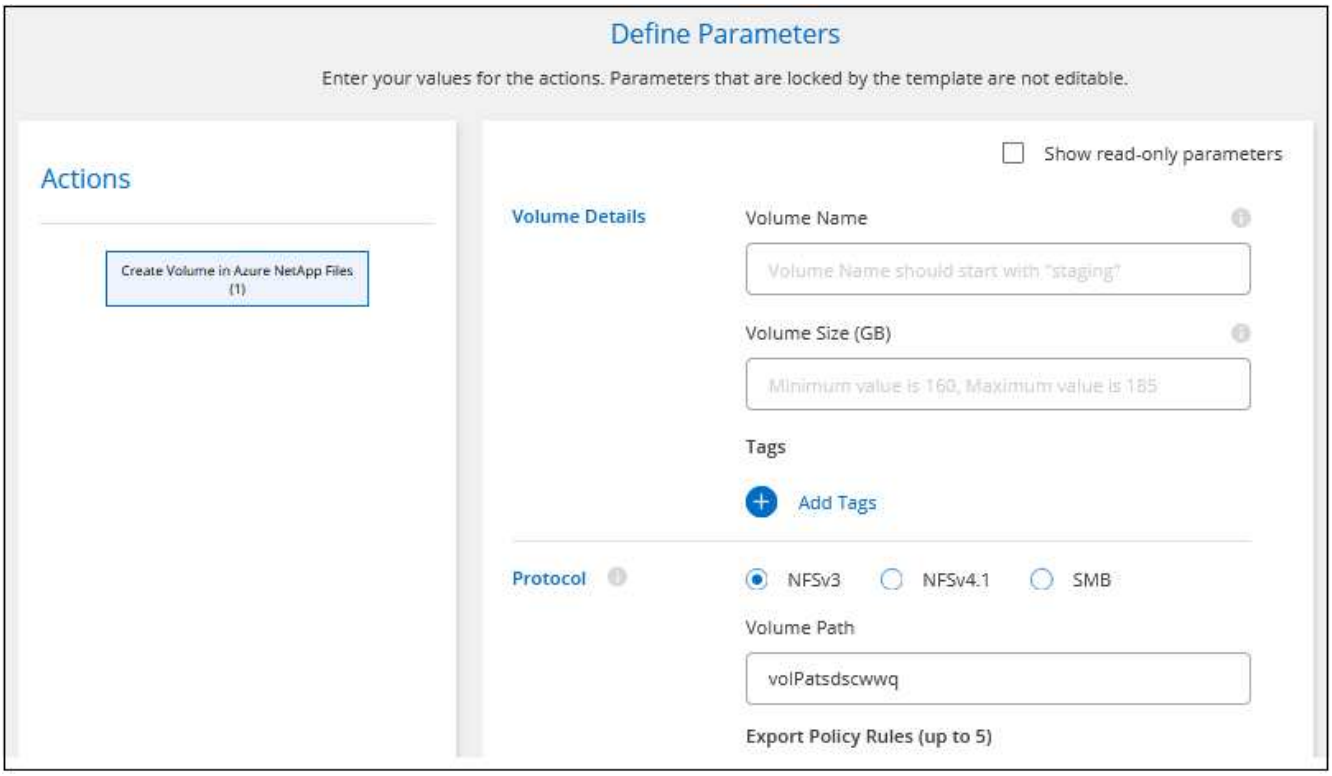
1. On the Canvas page, click the Azure NetApp Files working environment on which you want to provision a volume.
2. Click > **Add Volume From Template**.



3. In the *Select Template* page, select the template that you want to use to create the volume and click **Next**.



The *Define Parameters* page is displayed.



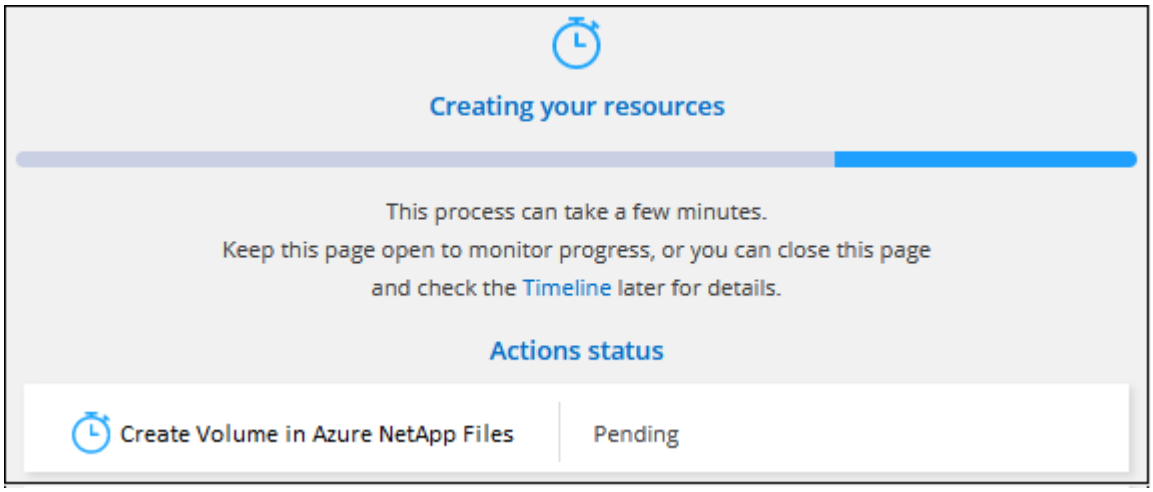
Note: You can click the checkbox **Show read-only parameters** to show all the fields that have been locked by the template if you want to see the values for those parameters. By default these predefined

fields are hidden and only the fields you need to complete are shown.

4. Add values for all of the parameters that are not hard-coded from the template. See create volumes for details about all the parameters you need to complete to deploy an Azure NetApp Files volume.
5. Click **Run Template** after you have defined all the parameters needed for this volume.

Result

BlueXP provisions the volume and displays a page so that you can see the progress.



Then the new volume is added to the working environment.

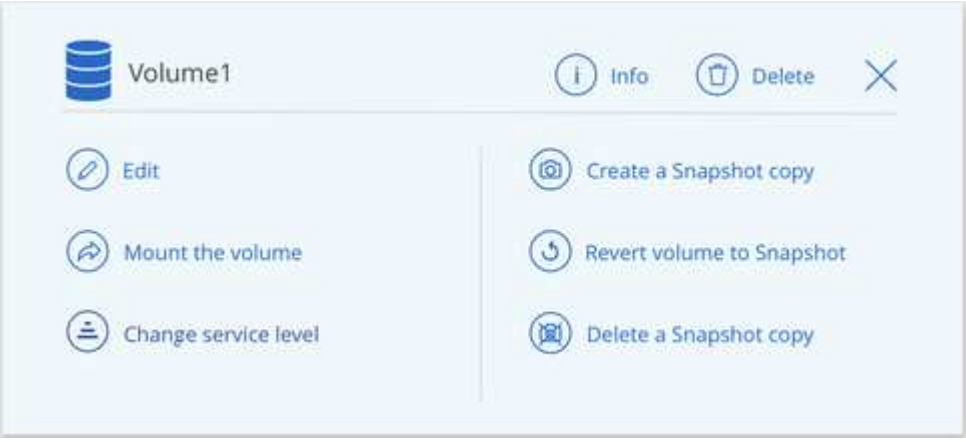
Continue with mounting the cloud volume.

Mount volumes

Access mounting instructions from within BlueXP so you can mount the volume to a host.

Steps

1. Open the working environment.
2. Hover over the volume and select **Mount the volume**.



3. Follow the instructions to mount the volume.

Manage existing volumes

You can manage existing volumes as your storage needs change. You can edit volumes, change a volume's service level, manage Snapshot copies, and delete volumes.

Edit a volume's size and tags

After you create a volume, you can modify its size and tags at any time.

Steps

1. Open the working environment.
2. Hover over the volume and select **Edit**.
3. Modify the size and tags as needed.
4. Click **Apply**.

Change the volume's service level

After you create a volume, you can change the service level at any time as long as the destination capacity pool already exists.

Steps

1. Open the working environment.
2. Hover over the volume and select **Change service level**.
3. Select the capacity pool that provides the service level that you want.
4. Click **Change**.

Result

The volume is moved to the other capacity pool with no impact to the volume.

Manage Snapshot copies

Snapshot copies provide a point-in-time copy of your volume. Create Snapshot copies, restore the data to a new volume, and delete Snapshot copies.

Steps

1. Open the working environment.
2. Hover over the volume and choose one of the available options to manage Snapshot copies:
   - **Create a Snapshot copy**
   - **Revert volume to Snapshot**
   - **Delete a Snapshot copy**
3. Follow the prompts to complete the selected action.

Delete volumes

Delete the volumes that you no longer need.

Steps

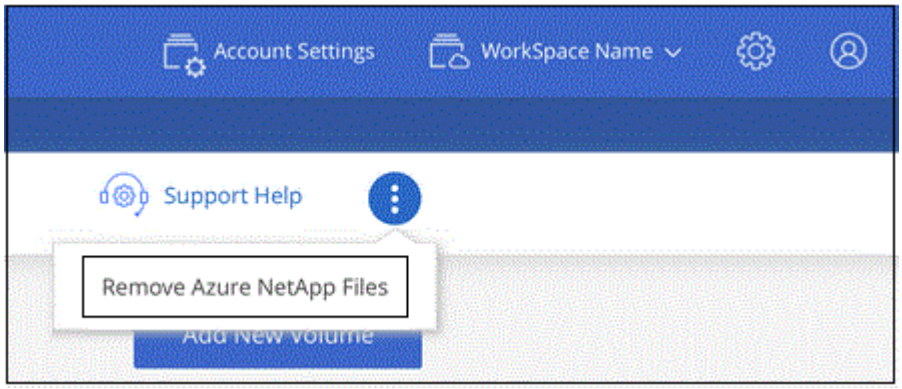
1. Open the working environment.
2. Hover over the volume and click **Delete**.
3. Confirm that you want to delete the volume.

Remove Azure NetApp Files

This action removes Azure NetApp Files from BlueXP. It doesn't delete your Azure NetApp Files account or volumes. You can add Azure NetApp Files back to BlueXP at any time.

Steps

1. Open the Azure NetApp Files working environment.
2. At the top right of the page, select the actions menu and click **Remove Azure NetApp Files**.



3. Click **Remove** to confirm.

Knowledge and support

Register for support

Support registration is required to receive technical support specific to BlueXP and its storage solutions and services. Support registration is also required to enable key workflows for Cloud Volumes ONTAP systems.

Registering for support does not enable NetApp support for a cloud provider file service. For technical support related to a cloud provider file service, its infrastructure, or any solution using the service, refer to "Getting help" in the BlueXP documentation for that product.

- Amazon FSx for ONTAP
- Azure NetApp Files
- Cloud Volumes Service for Google Cloud

Support registration overview

There are two forms of registration to activate support entitlement:

- Registering your BlueXP account ID support subscription (your 20 digit 960xxxxxxxxx serial number located on the Support Resources page in BlueXP).

  This serves as your single support subscription ID for any service within BlueXP. Each BlueXP account-level support subscription must be registered.

- Registering the Cloud Volumes ONTAP serial numbers associated with a subscription in your cloud provider's marketplace (these are 20 digit 909201xxxxxxxx serial numbers).

  These serial numbers are commonly referred to as *PAYGO serial numbers* and get generated by BlueXP at the time of Cloud Volumes ONTAP deployment.

Registering both types of serial numbers enables capabilities like opening support tickets and automatic case generation. Registration is completed by adding NetApp Support Site (NSS) accounts to BlueXP as described below.

Register your BlueXP account for NetApp support

To register for support and activate support entitlement, one user in your BlueXP account must associate a NetApp Support Site account with their BlueXP login. How you register for NetApp support depends on whether you already have a NetApp Support Site (NSS) account.

Existing customer with an NSS account

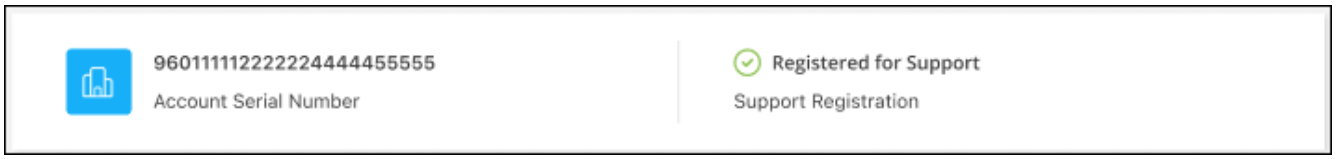
If you're a NetApp customer with an NSS account, you simply need to register for support through BlueXP.

Steps

1. In the upper right of the BlueXP console, select the Settings icon, and select **Credentials**.
2. Select **User Credentials**.

3. Select **Add NSS credentials** and follow the NetApp Support Site (NSS) Authentication prompt.
4. To confirm that the registration process was successful, select the Help icon, and select **Support**.

   The **Resources** page should show that your account is registered for support.



   Note that other BlueXP users will not see this same support registration status if they have not associated a NetApp Support Site account with their BlueXP login. However, that doesn't mean that your BlueXP account is not registered for support. As long as one user in the account has followed these steps, then your account has been registered.

Existing customer but no NSS account

If you're an existing NetApp customer with existing licenses and serial numbers but *no* NSS account, you need to create an NSS account and associate it with your BlueXP login.

Steps

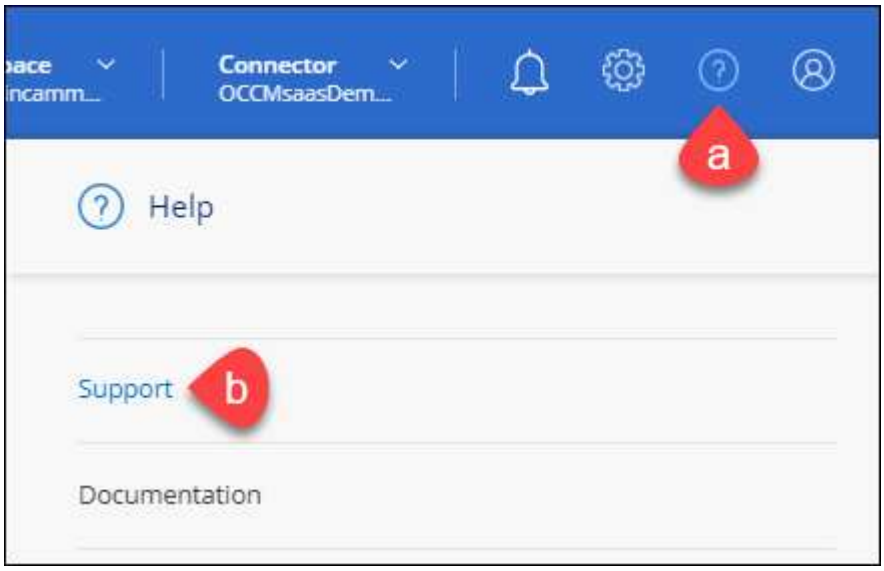
1. Create a NetApp Support Site account by completing the NetApp Support Site User Registration form
   a. Be sure to select the appropriate User Level, which is typically **NetApp Customer/End User**.
   b. Be sure to copy the BlueXP account serial number (960xxxx) used above for the serial number field. This will speed up the account processing.
2. Associate your new NSS account with your BlueXP login by completing the steps under Existing customer with an NSS account.

Brand new to NetApp

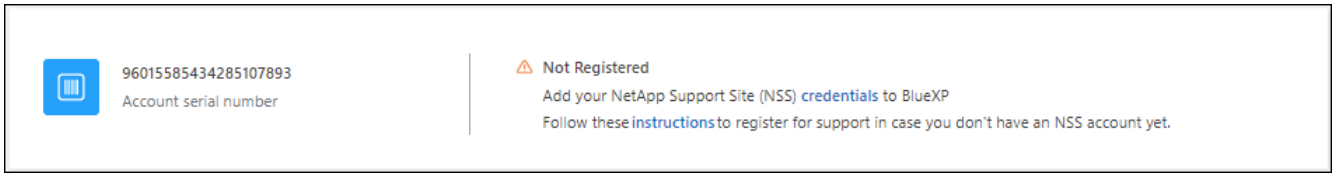
If you are brand new to NetApp and you don't have an NSS account, follow each step below.

Steps

1. In the upper right of the BlueXP console, select the Help icon, and select **Support**.



2. Locate your account ID serial number from the Support Registration page.



3. Navigate to NetApp's support registration site and select **I am not a registered NetApp Customer**.
4. Fill out the mandatory fields (those with red asterisks).
5. In the **Product Line** field, select **Cloud Manager** and then select your applicable billing provider.
6. Copy your account serial number from step 2 above, complete the security check, and then confirm that you read NetApp's Global Data Privacy Policy.

   An email is immediately sent to the mailbox provided to finalize this secure transaction. Be sure to check your spam folders if the validation email doesn't arrive in few minutes.

7. Confirm the action from within the email.

   Confirming submits your request to NetApp and recommends that you create a NetApp Support Site account.

8. Create a NetApp Support Site account by completing the NetApp Support Site User Registration form
   a. Be sure to select the appropriate User Level, which is typically **NetApp Customer/End User**.
   b. Be sure to copy the account serial number (960xxxx) used above for the serial number field. This will speed up the account processing.

After you finish

NetApp should reach out to you during this process. This is a one-time onboarding exercise for new users.

Once you have your NetApp Support Site account, associate the account with your BlueXP login by completing the steps under Existing customer with an NSS account.

Associate NSS credentials for Cloud Volumes ONTAP support

Associating NetApp Support Site credentials with your BlueXP account is required to enable the following key workflows for Cloud Volumes ONTAP:

- Registering pay-as-you-go Cloud Volumes ONTAP systems for support

  Providing your NSS account is required to activate support for your system and to gain access to NetApp technical support resources.

- Deploying Cloud Volumes ONTAP when you bring your own license (BYOL)

  Providing your NSS account is required so that BlueXP can upload your license key and to enable the subscription for the term that you purchased. This includes automatic updates for term renewals.

- Upgrading Cloud Volumes ONTAP software to the latest release

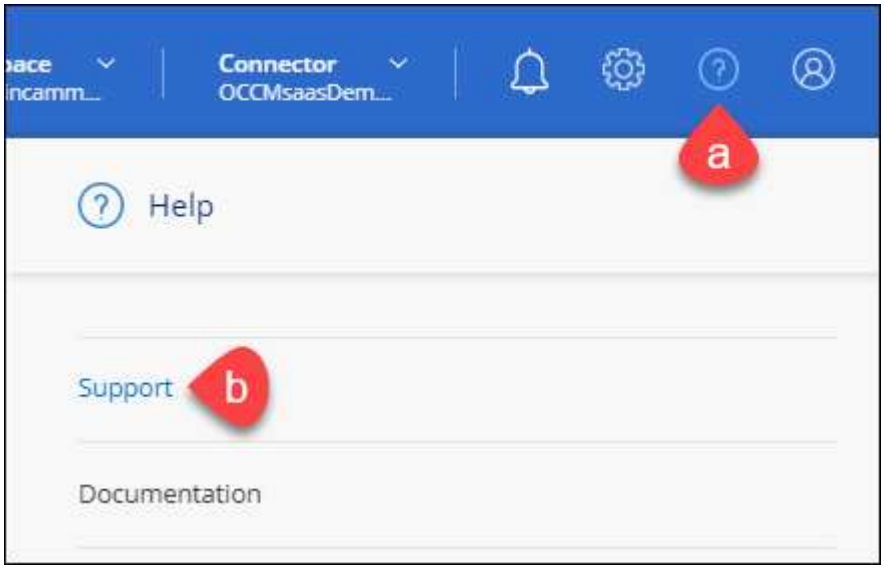
Associating NSS credentials with your BlueXP account is different than the NSS account that is associated with a BlueXP user login.

These NSS credentials are associated with your specific BlueXP account ID. Users who belong to the BlueXP account can access these credentials from **Support > NSS Management**.

- If you have a customer-level account, you can add one or more NSS accounts.
- If you have a partner or reseller account, you can add one or more NSS accounts, but they can't be added alongside customer-level accounts.

Steps

1. In the upper right of the BlueXP console, select the Help icon, and select **Support**.



2. Select **NSS Management > Add NSS Account**.
3. When you're prompted, select **Continue** to be redirected to a Microsoft login page.

   NetApp uses Microsoft Entra ID as the identity provider for authentication services specific to support and licensing.

4. At the login page, provide your NetApp Support Site registered email address and password to perform the authentication process.

   These actions enable BlueXP to use your NSS account for things like license downloads, software upgrade verification, and future support registrations.

   Note the following:

   - The NSS account must be a customer-level account (not a guest or temp account). You can have multiple customer-level NSS accounts.
   - There can be only one NSS account if that account is a partner-level account. If you try to add customer-level NSS accounts and a partner-level account exists, you'll get the following error message:

     "The NSS customer type is not allowed for this account as there are already NSS Users of different type."

     The same is true if you have pre-existing customer-level NSS accounts and try to add a partner-level account.

   - Upon successful login, NetApp will store the NSS user name.

This is a system-generated ID that maps to your email. On the **NSS Management** page, you can display your email from the ••• menu.

- If you ever need to refresh your login credential tokens, there is also an **Update Credentials** option in the ••• menu.

  Using this option prompts you to log in again. Note that the token for these accounts expire after 90 days. A notification will be posted to alert you of this.

Get help

NetApp provides support for BlueXP and its cloud services in a variety of ways. Extensive free self-support options are available 24x7, such as knowledgebase (KB) articles and a community forum. Your support registration includes remote technical support via web ticketing.

Get support for a cloud provider file service

For technical support related to a cloud provider file service, its infrastructure, or any solution using the service, refer to "Getting help" in the BlueXP documentation for that product.

- Amazon FSx for ONTAP
- Azure NetApp Files
- Cloud Volumes Service for Google Cloud

To receive technical support specific to BlueXP and its storage solutions and services, use the support options described below.

Use self-support options

These options are available for free, 24 hours a day, 7 days a week:

- Documentation

  The BlueXP documentation that you're currently viewing.

- Knowledge base

  Search through the BlueXP knowledge base to find helpful articles to troubleshoot issues.

- Communities

  Join the BlueXP community to follow ongoing discussions or create new ones.

Create a case with NetApp support

In addition to the self-support options above, you can work with a NetApp Support specialist to resolve any issues after you activate support.

Before you get started

- To use the **Create a Case** capability, you must first associate your NetApp Support Site credentials with your BlueXP login. Learn how to manage credentials associated with your BlueXP login.
- If you're opening a case for an ONTAP system that has a serial number, then your NSS account must be associated with the serial number for that system.

Steps

1. In BlueXP, select **Help > Support**.
2. On the **Resources** page, choose one of the available options under Technical Support:

   a. Select **Call Us** if you'd like to speak with someone on the phone. You'll be directed to a page on netapp.com that lists the phone numbers that you can call.

   b. Select **Create a Case** to open a ticket with a NetApp Support specialist:

      - **Service**: Select the service that the issue is associated with. For example, BlueXP when specific to a technical support issue with workflows or functionality within the service.
      - **Working Environment**: If applicable to storage, select **Cloud Volumes ONTAP** or **On-Prem** and then the associated working environment.

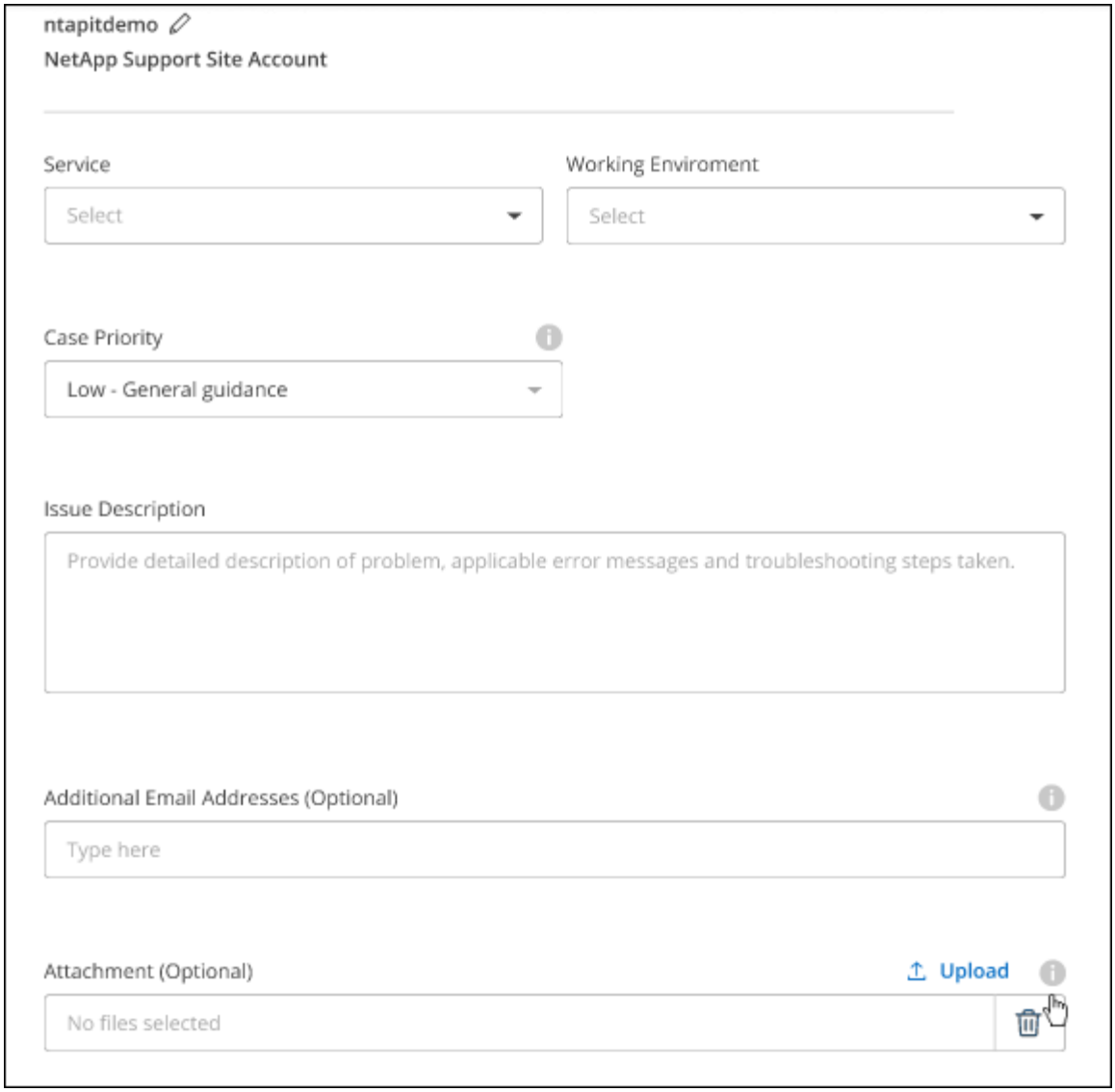
        The list of working environments are within scope of the BlueXP account, workspace, and Connector you have selected in the top banner of the service.

      - **Case Priority**: Choose the priority for the case, which can be Low, Medium, High, or Critical.

        To learn more details about these priorities, hover your mouse over the information icon next to the field name.

      - **Issue Description**: Provide a detailed description of your problem, including any applicable error messages or troubleshooting steps that you performed.
      - **Additional Email Addresses**: Enter additional email addresses if you'd like to make someone else aware of this issue.
      - **Attachment (Optional)**: Upload up to five attachments, one at a time.

        Attachments are limited to 25 MB per file. The following file extensions are supported: txt, log, pdf, jpg/jpeg, rtf, doc/docx, xls/xlsx, and csv.



After you finish

A pop-up will appear with your support case number. A NetApp Support specialist will review your case and get back to you soon.

For a history of your support cases, you can select **Settings > Timeline** and look for actions named "create support case." A button to the far right lets you expand the action to see details.

It's possible that you might encounter the following error message when trying to create a case:

"You are not authorized to Create a Case against the selected service"

This error could mean that the NSS account and the company of record it's associated with is not the same company of record for the BlueXP account serial number (ie. 960xxxx) or the working environment serial number. You can seek assistance using one of the following options:

- Use the in-product chat
- Submit a non-technical case at https://mysupport.netapp.com/site/help

Manage your support cases (Preview)

You can view and manage active and resolved support cases directly from BlueXP. You can manage the cases associated with your NSS account and with your company.

Case management is available as a Preview. We plan to refine this experience and add enhancements in upcoming releases. Please send us feedback by using the in-product chat.

Note the following:

- The case management dashboard at the top of the page offers two views:
  - The view on the left shows the total cases opened in the past 3 months by the user NSS account you provided.
  - The view on the right shows the total cases opened in the past 3 months at your company level based on your user NSS account.

  The results in the table reflect the cases related to the view that you selected.

- You can add or remove columns of interest and you can filter the contents of columns like Priority and Status. Other columns provide just sorting capabilities.

  View the steps below for more details.

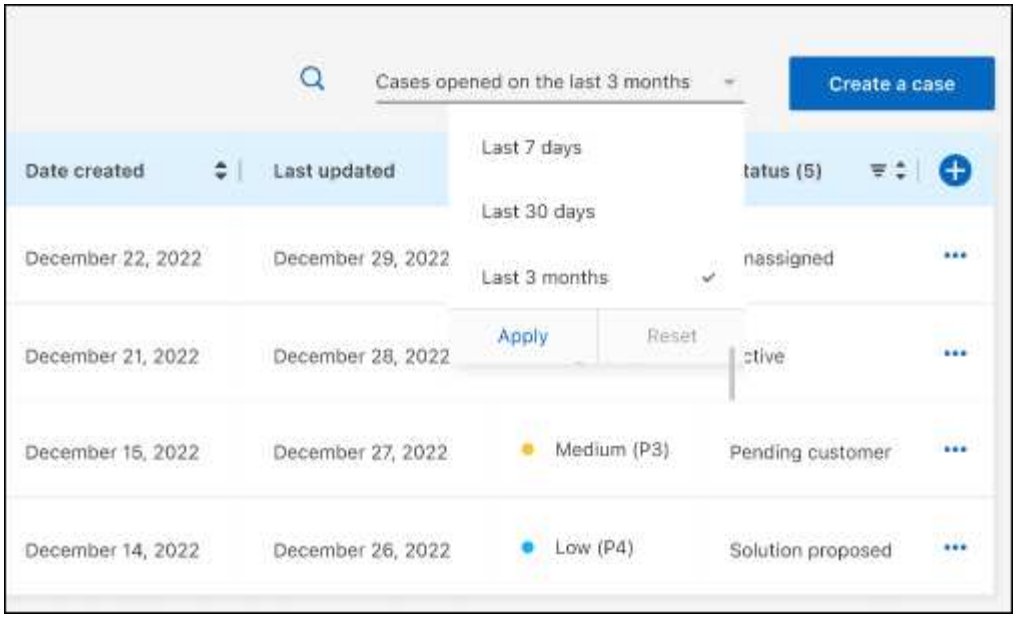
- At a per-case level, we offer the ability to update case notes or close a case that is not already in Closed or Pending Closed status.

Steps

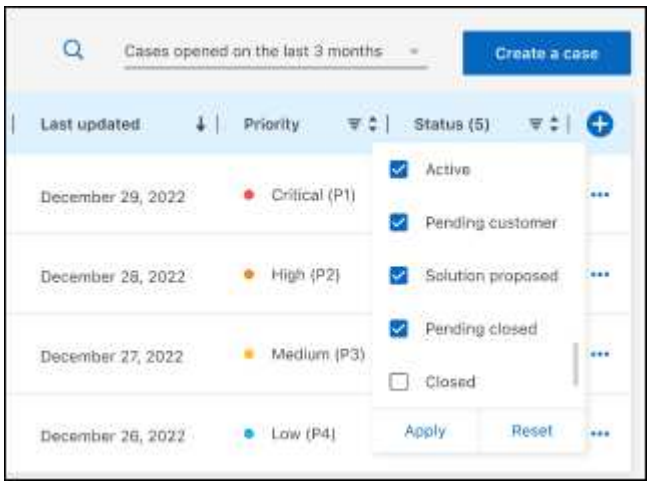
1. In BlueXP, select **Help > Support**.
2. Select **Case Management** and if you're prompted, add your NSS account to BlueXP.

   The **Case management** page shows open cases related to the NSS account that is associated with your BlueXP user account. This is the same NSS account that appears at the top of the **NSS management** page.

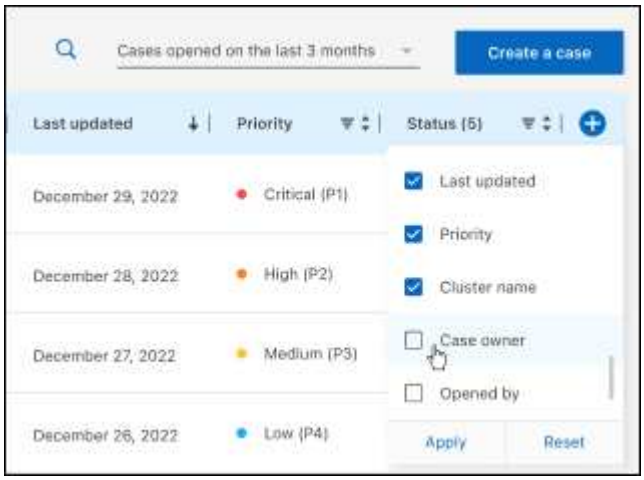
3. Optionally modify the information that displays in the table:
   - Under **Organization's cases**, select **View** to view all cases associated with your company.
   - Modify the date range by choosing an exact date range or by choosing a different time frame.



- Filter the contents of the columns.



- Change the columns that appear in the table by selecting + and then choosing the columns that you'd like to display.

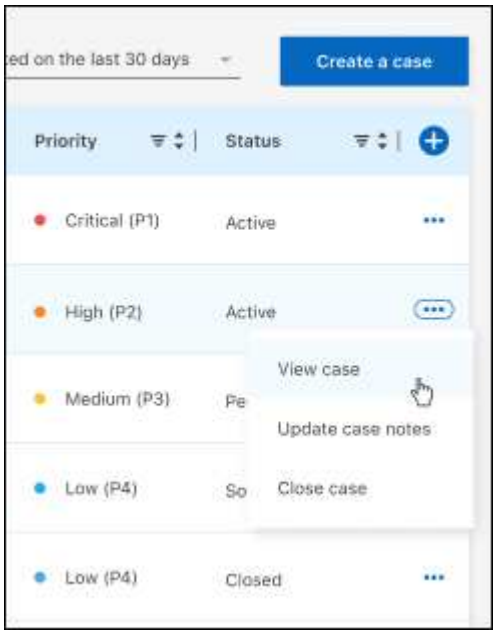


4. Manage an existing case by selecting ••• and selecting one of the available options:

   - **View case**: View full details about a specific case.
   - **Update case notes**: Provide additional details about your problem or select **Upload files** to attach up to a maximum of five files.

     Attachments are limited to 25 MB per file. The following file extensions are supported: txt, log, pdf, jpg/jpeg, rtf, doc/docx, xls/xlsx, and csv.

   - **Close case**: Provide details about why you're closing the case and select **Close case**.



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